

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: CSA0714

COURSE NAME: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Sri Janani B(192421036) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sri Janani B

Annexure A

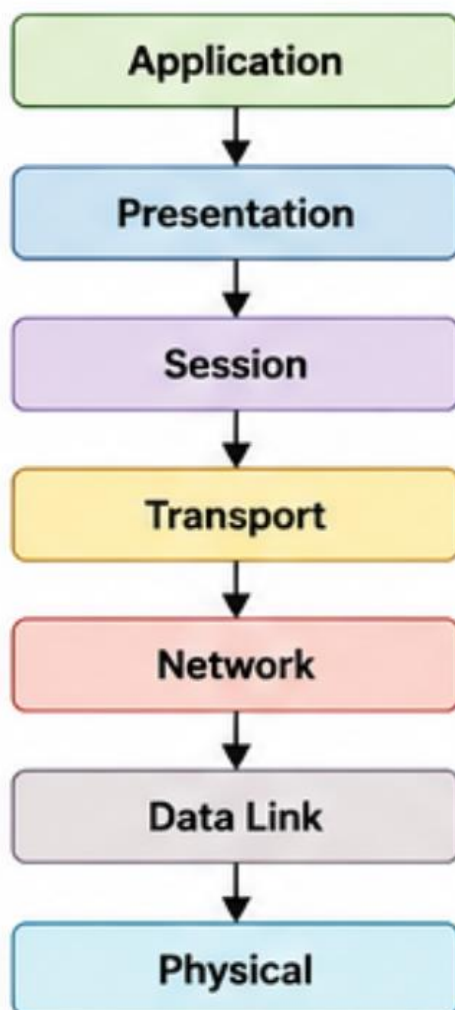
CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

1. Missing OSI Layers



Missing Layers:

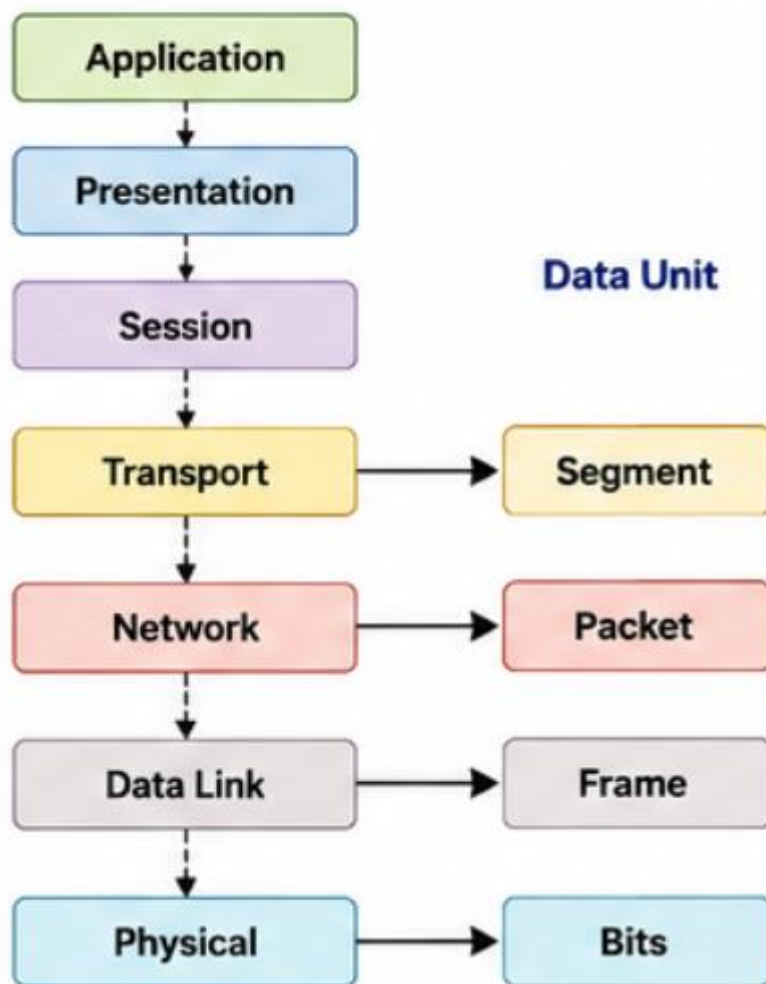
- Session Layer
- Data Link Layer

Explanation:

- **Session Layer:**
Establishes, manages and terminates communication sessions. It also provides synchronization and recovery.
- **Data Link Layer:**
Performs framing, MAC addressing, error detection and reliable delivery between two adjacent nodes.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

2. Data Units and OSI Layers



Explanation:

- **Transport Layer** divides data into Segments.
- **Network Layer** adds IP header and creates Packets.
- **Data Link Layer** adds MAC header and trailer to form Frames.
- **Physical Layer** converts frames into Bits and sends over the medium.

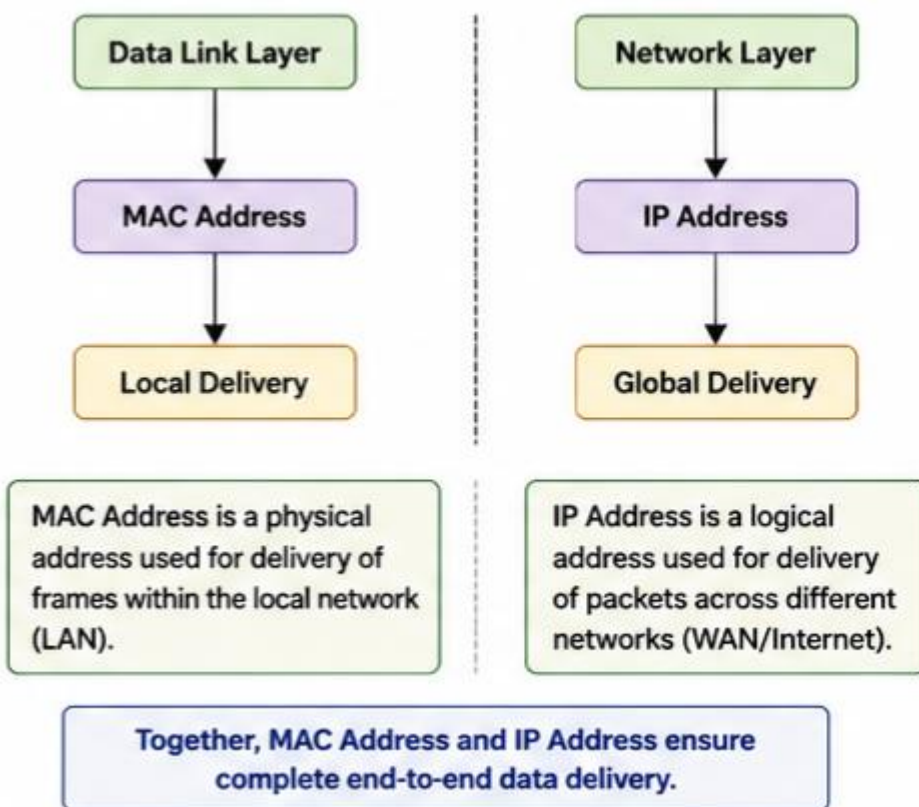
3. Given the concept map:

DataLink → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

3. MAC Address and IP Address



4. A concept map shows:

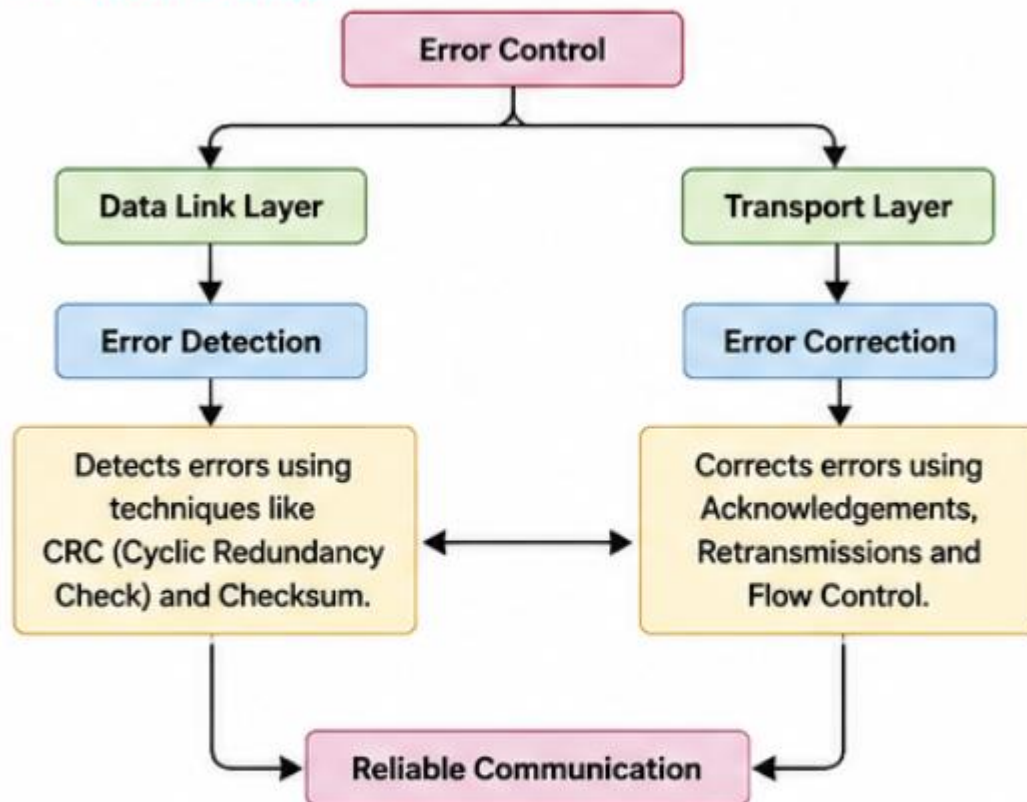
ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

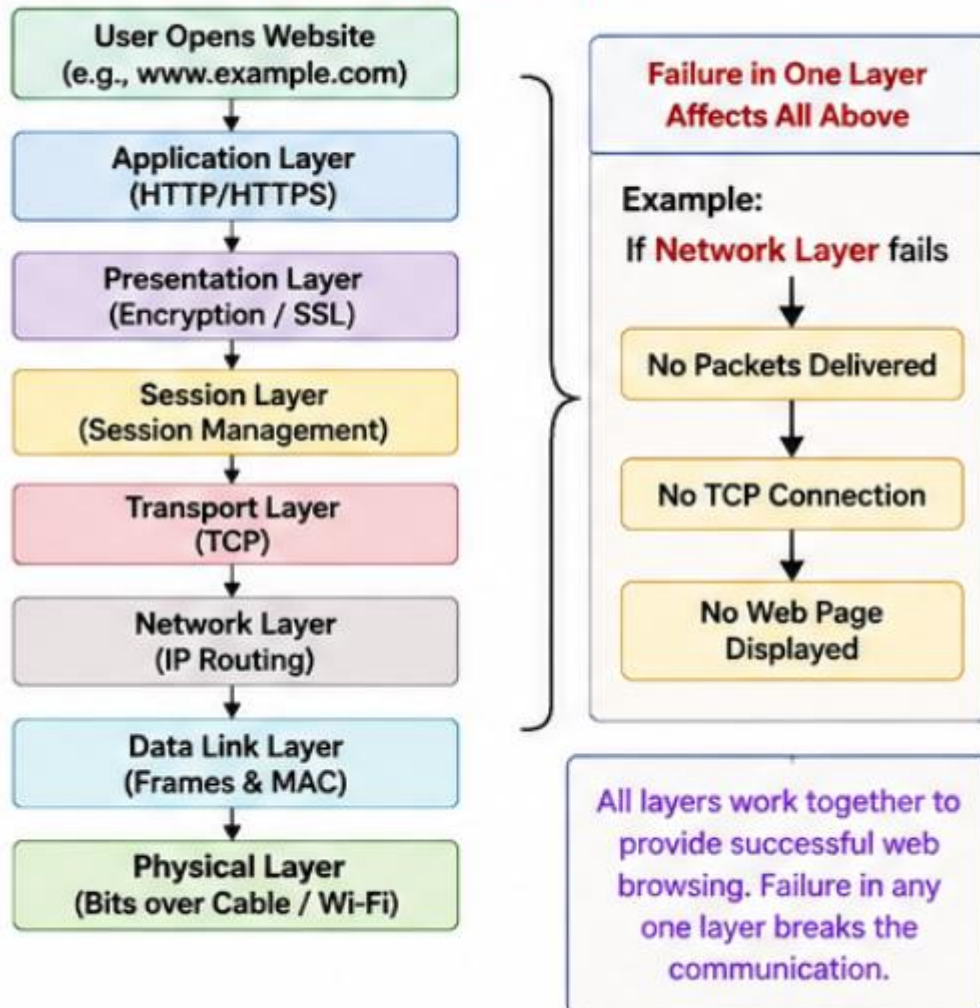
4. Error Control



Data Link Layer detects errors, Transport Layer corrects them. Their interaction increases reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

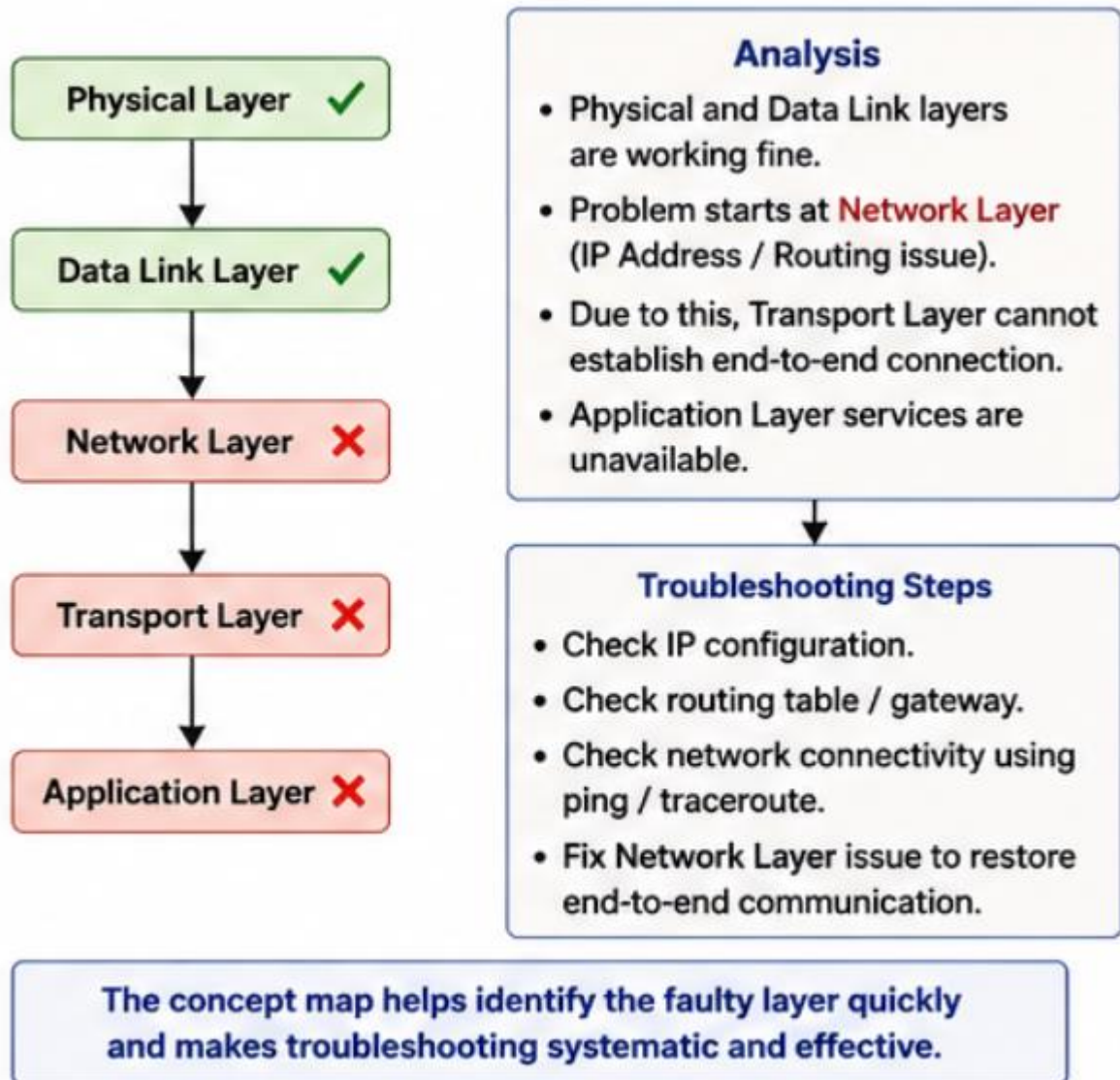
5. Web Browsing Using OSI Layers



6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗ Analyze the problem and explain how the concept map helps in troubleshooting.

6. Troubleshooting Concept Map



Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation